

Notes: Chan, Menkveld, & Yang (JF, 2008)

Information Asymmetry and Asset Prices

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1 Big picture

- **Question.** Does information asymmetry between domestic and foreign investors affect asset prices?
- **Empirical laboratory.** China before March 2001 had segmented share classes: domestic investors traded A-shares, foreign investors traded B-shares. The two claims had the same voting and cash-flow rights, but B-shares traded at large discounts.
- **Key puzzle.** Foreign investors are internationally diversified, so one might expect a foreign-share premium. Instead, China displayed a foreign-share discount: B-shares were priced below A-shares.
- **Main idea.** Domestic A-share investors are more likely to possess private information about Chinese firms. Foreign B-share investors are less informed and therefore value the same cash-flow rights less.
- **Contribution.** The paper imports market microstructure measures of information asymmetry into an asset-pricing question. It estimates price impact, the adverse selection component of the bid-ask spread, and PIN for each firm's A- and B-share markets.
- **Main evidence.** On a univariate basis, the A-minus-B difference in price impact explains about 44% of the cross-sectional variation in B-share discounts, and the A-minus-B difference in the adverse selection spread component explains about 46%.
- **Multivariate result.** Price impact and adverse selection remain far more statistically important than controls for turnover, number of trades, market capitalization, and momentum.
- **Policy shock.** In March 2001, China allowed domestic investors to trade B-shares with foreign currency. The average B-share discount fell from about 72% to about 43%, and the information-asymmetry difference between markets narrowed.
- **Economic takeaway.** Information asymmetry is not just a trading frictions story. It is reflected in cross-sectional asset prices when otherwise identical claims are held by investors with different information sets.

2 Data and measurement

2.1 A- and B-share market structure

- A-shares are domestic ordinary shares, denominated and traded in yuan by Chinese citizens.
- B-shares are ordinary shares offered to foreign investors. They have the same voting and dividend rights as A-shares but are traded in foreign currency.
- Before March 2001, A- and B-share markets were fully segmented: domestic investors could not trade B-shares and foreigners could not trade A-shares.
- Since March 2001, domestic investors have been permitted to trade B-shares using foreign currency.
- Both Shanghai and Shenzhen were order-driven markets with electronic limit order books and no designated market makers.

Interpretation: The setting is unusually clean because the A- and B-share claims have identical cash-flow and control rights. Therefore, price differences are easier to interpret as coming from market segmentation, investor clienteles, information, liquidity, or speculative forces.

2.2 Sample construction

- The data contain all time-stamped trades and quotes from January 2000 to November 2001 for stocks traded on the Shanghai and Shenzhen exchanges.

- The main pre-opening sample is January to December 2000, when both markets were fully segmented.
- The sample is restricted to firms with both A- and B-shares trading throughout the sample period.
- After excluding firms with little or no B-share volume, the final main sample contains 76 firms.
- The authors remove days with trading halts or very limited trading, stale quotes with zero depth, and the first and last 15 minutes of each session.

Interpretation: The matched-share design lets the authors compare domestic and foreign markets for the same firm rather than compare different firms with different fundamentals.

2.3 Foreign share discount

The main asset-pricing outcome is the foreign share discount:

$$FSD_i = 1 - \frac{P_{B,i}}{P_{A,i}}, \quad (1)$$

where $P_{A,i}$ is the A-share price and $P_{B,i}$ is the B-share price, converted into yuan using the official exchange rate.

Interpretation: A larger FSD_i means that the foreign B-share trades at a larger discount relative to the local A-share of the same firm.

2.4 Information asymmetry measures

The paper estimates three microstructure-based measures.

- **Price impact (PI).** Higher price impact means a signed trade moves prices more, which is interpreted as stronger informed trading or lower market depth.
- **Adverse selection spread component (AS).** This decomposes the spread into components and extracts the part associated with adverse selection from informed traders.
- **Probability of informed trading (PIN).** This estimates the probability that a randomly selected trade comes from informed traders using the Easley-Kiefer-O'Hara framework.

Interpretation: The empirical tests use differences across markets, such as $PI_A - PI_B$ and $AS_A - AS_B$, because the hypothesis is that domestic investors are more informed than foreign investors.

2.5 Control variables

The multivariate regressions include controls for alternative explanations:

- turnover in A- and B-share markets;
- number of trades in A- and B-share markets;
- market capitalization of A- and B-shares;
- momentum, measured using A-share returns.

Interpretation: These controls address competing stories: liquidity/trading activity, speculative trading, firm size and disclosure, and differential risk or momentum.

3 Models and empirical specifications

3.1 Theoretical mechanism

The model extends Grossman and Stiglitz-style noisy rational expectations to two segmented markets.

- Domestic A-share investors contain informed and uninformed investors.
- The fraction of informed A-share investors is λ .
- Foreign B-share investors are assumed uninformed for convenience.
- Uninformed investors can infer information imperfectly from prices.

Interpretation: As λ rises, the domestic market contains more private information. This affects both A- and B-share prices because B-share investors learn indirectly from the A-share price.

3.2 Price impact regression

Let P_t be the transaction price, $Q_t \in \{+1, -1\}$ the trade sign, and V_t trade size. The price impact model is

$$P_t - P_{t-1} = \gamma Q_t V_t + \phi(Q_t - Q_{t-1}) + e_t. \quad (2)$$

- γ is the price impact coefficient, or inverse market depth.
- ϕ captures the gross profit component associated with order processing and inventory.
- The paper estimates this separately for A- and B-shares.

Interpretation: If $\gamma_A > \gamma_B$, then the same signed order flow has a larger price effect in the A-share market. In the model, this can arise because the A-share market contains more informed domestic investors.

3.3 Adverse selection spread decomposition

The paper lets the adverse selection and gross profit components vary with trade size:

$$\gamma_t = z_0 + z_1 V_t, \quad \phi_t = c_0 + c_1 V_t. \quad (3)$$

The estimating equation becomes

$$P_t - P_{t-1} = c_0(Q_t - Q_{t-1}) + c_1(Q_t V_t - Q_{t-1} V_{t-1}) + z_0 Q_t + z_1 \hat{\varepsilon}_t + u_t, \quad (4)$$

where $\hat{\varepsilon}_t$ is the innovation in signed order flow. The adverse selection measure is

$$AS_i = z_0 + z_1 V_i^*, \quad (5)$$

where V_i^* is the median order size.

Interpretation: This measure tries to isolate the part of trading costs that arises because liquidity suppliers face informed traders, rather than the part due to ordinary order processing.

3.4 PIN model

The PIN model assumes information events arrive with probability α , bad news with probability δ , informed orders at rate μ , and uninformed buy and sell orders at rates ε_b and ε_s .

The probability of informed trading is

$$PIN = \frac{\alpha\mu}{\alpha\mu + \varepsilon_b + \varepsilon_s}. \quad (6)$$

Interpretation: PIN can be high either because informed trading is frequent or because uninformed trading is scarce. This matters in this paper: the B-share market can have higher PIN despite lower total informed trading because uninformed trading is also low.

3.5 Foreign share discount regressions

The main cross-sectional specification is

$$FSD_i = a + bIA_i + \Gamma'X_i + u_i, \quad (7)$$

where IA_i is one or more information-asymmetry differences, such as

$$PI_A - PI_B, \quad AS_A - AS_B, \quad PIN_A - PIN_B, \quad (8)$$

and X_i contains turnover, number of trades, market capitalization, and momentum.

Interpretation: A positive b means that firms with greater A-minus-B information asymmetry have larger B-share discounts.

3.6 Post-opening analysis

The March 2001 policy change allows domestic investors to enter the B-share market. The paper compares pre- and post-opening microstructure measures and repeats the discount regressions after the opening.

Interpretation: If information asymmetry drives the discount, then allowing domestic investors into the B-share market should reduce the information gap and reduce the discount.

4 Empirical design and interpretation

4.1 Why the design is informative

- A- and B-shares represent the same firm rights, so fundamentals are held fixed at the firm level.
- The two markets are segmented before 2001, making investor clientele differences sharp.
- Transaction data allow the authors to estimate information asymmetry directly, rather than proxy for it with size or disclosure.
- The 2001 market opening provides a useful institutional change that should reduce the B-share information disadvantage.

Interpretation: The paper is not a randomized experiment, but the matched-share setting and policy change make the information-asymmetry interpretation more credible than a purely cross-sectional firm-size test.

4.2 Competing explanations

- **Trading activity/liquidity:** B-shares trade less actively, so investors may require a discount.
- **Speculative bubbles:** A-share prices may be high because domestic investors speculate more aggressively.
- **Differential risk:** Domestic and foreign investors face different opportunity sets and therefore value the same cash flows differently.
- **Firm size/disclosure:** Larger firms may have lower information costs and smaller discounts.

Interpretation: The central empirical question is whether microstructure information measures remain important after these competing variables are included.

4.3 Limits of the interpretation

- The measures capture informed trading, but not whether the information is fundamental, rumor-based, or linked to price manipulation.
- Larger A-share price impact could partly reflect informed traders' ability to hide in a more active market.
- The cross-sectional regressions can show strong association, but they cannot fully isolate all sources of segmentation or investor demand.

Interpretation: The paper's strongest claim is that information asymmetry is priced; it does not require every piece of private information to be fundamental value information.

5 Tables and figures

5.1 Figures 1-3: theoretical mechanism

Read:

- Figure 1 shows that both P_A and P_B increase with the share of informed domestic investors, but P_A is higher than P_B .
- The A-minus-B price gap first rises and then declines as λ increases.
- Figure 2 shows that A-market price impact can rise and then fall with λ , while B-market price impact declines.
- Figure 3 connects the theory to the empirical test: for most parameter values, the B-share discount is positively related to the A-minus-B price impact difference.

Economic message: The model explains why a microstructure object, relative price impact, should be linked to an asset-pricing object, the B-share discount.

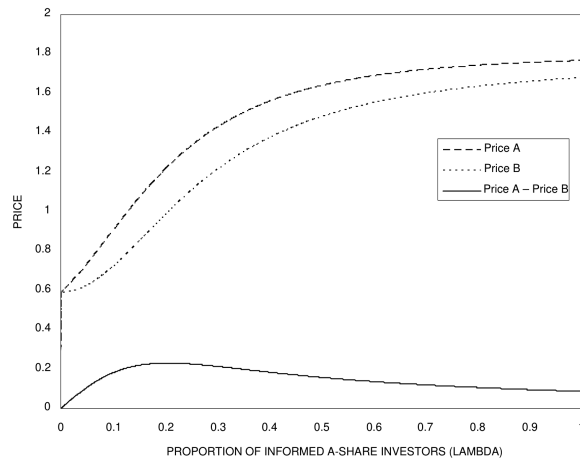


Figure 1: prices and foreign share discount as informed A-share investors rise

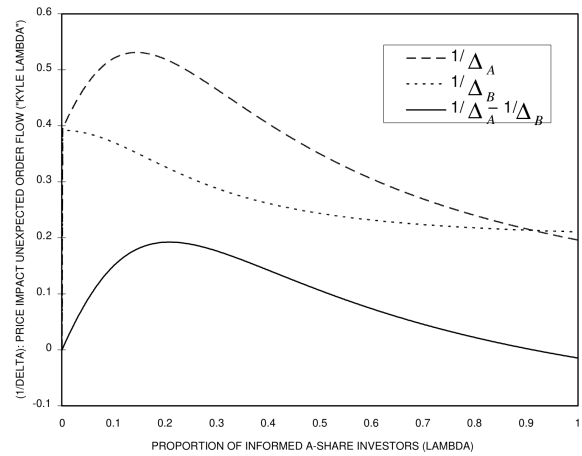


Figure 2: price impact and market depth as informed A-share investors rise

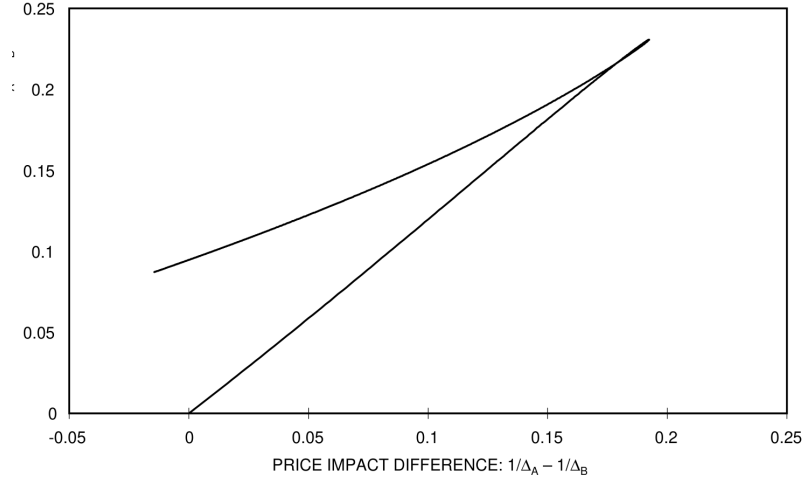


Figure 3: B-share discount versus price-impact difference

5.2 Table I: trading summary statistics

Read:

- The average A-share transaction price is 14.29 yuan, while the average B-share price is 3.10 yuan.
- B-shares trade much less frequently: average time between trades is 313 seconds for B-shares and 62 seconds for A-shares.
- B-share daily volume is about half the A-share volume: 854,000 versus 1,684,000 shares per day.
- B-share trade size is larger: 16,769 shares versus 5,769 shares.
- Relative spreads are much larger in B-shares: 128 bps quoted spread versus 18 bps for A-shares; 126 bps effective spread versus 24 bps for A-shares.
- B-share displayed depth is larger despite lower volume, which warns against equating volume with liquidity.

Economic message: The raw trading environment already shows strong segmentation. B-shares are cheaper, less frequently traded, and have much higher relative spreads.

Table I
Summary Statistics for A- and B-Share Trading

This table presents summary statistics of trading in Chinese A- (local) and B- (foreign) shares for January through December 2000, when both markets were fully separated. We provide cross-sectional statistics on trading variables for 76 firms whose shares were listed both as A- and B-shares. To facilitate comparisons across markets, for each share we only include those days for which there was uninterrupted trading in both the A- and B-share markets.

	A- (Local) Share				B- (Foreign) Share			
	Mean	σ	Min	Max	Mean	σ	Min	Max
Avg Transaction Price (Yuan)	14.29	4.17	6.15	26.05	3.10	1.14	1.21	7.96
St. Dev. of Price Change ^a (Yuan)	0.03	0.04	0.01	0.10	0.03	0.04	0.01	0.11
Avg Time between Trades (sec)	62	20	30	112	313	91	127	546
Avg Daily Volume (1,000 shares)	1,684	1,321	274	7,018	854	653	95	3,014
Avg Trade Size (shares)	5,769	3,011	1,957	18,536	16,769	8,842	4,081	40,773
Avg Quoted Spread ^b (Yuan)	0.027	0.016	0.011	0.124	0.035	0.014	0.013	0.085
Avg Relative Quoted Spread ^b (bps)	18	6	10	46	128	42	46	246
Avg Effective Spread ^b (Yuan)	0.035	0.019	0.013	0.135	0.035	0.012	0.013	0.077
Avg Relative Effective Spread ^b (bps)	24	6	14	50	126	40	52	272
Avg Depth at the Best Quote ^{bc} (shares)	4,679	2,976	1,275	18,061	26,720	20,194	5,868	95,360
Number of Stocks	76				76			

^aBased on midquote changes in transaction time.

^bTime-weighted averages.

^cAverage of the best bid depth and best ask depth.

Table I: summary statistics for A- and B-share trading

5.3 Table II: estimating PI, AS, and PIN

Read:

- Panel A shows that the average price impact estimate is much larger in A-shares than B-shares, roughly 9.66 versus 2.61.
- Panel B shows a larger adverse selection spread component in A-shares than B-shares for median-size trades.
- Panel C shows that the arrival rate of informed traders is higher in A-shares, but PIN can be higher in B-shares because uninformed B-share trading is also much lower.
- The table demonstrates why the paper uses several information measures rather than a single proxy.

Economic message: The A-share market exhibits stronger informed-trading effects even though it is more active. This is the core microstructure fact behind the pricing test.

Table II
Information Asymmetry: PI, AS Component of the Spread, and PIN

This table presents cross-sectional statistics on the parameter estimates of three econometric models that provide us with measures on the level of information asymmetry in the market, namely, the price impact measure (PI), the adverse selection (AS) component of the spread, and the probability of informed trading (PIN). The three econometric models are estimated based on (local) A-share and (foreign) B-share transaction data for 76 stocks from January through December 2000. Panel A presents the estimates of the PI, which are based on Kyle (1985) and Glosten and Harris (1988)

$$P_t - P_{t-1} = \gamma Q_t V_t + \phi(Q_t - Q_{t-1}) + \epsilon_t,$$

where P_t is the price of transaction t , Q_t is an indicator that classifies the transaction as buyer-initiated (+1) or seller initiated (-1), V_t is the transaction size, ϵ_t is the unobservable error term, ϕ is the gross profit component associated with an order, and γ is the (inverse) market depth parameter, which is directly related to the price impact parameters introduced in the theoretical model. Panel B presents the estimates of a spread decomposition model based on Glosten and Harris (1988), defined as

$$P_t - P_{t-1} = c_0(Q_t - Q_{t-1}) + c_1(Q_t V_t - Q_{t-1} V_{t-1}) + c_2 Q_t + c_3 V_t + u_t \text{ with } u_t = \eta_t - \gamma \eta_{t-1},$$

where η_t is the innovation in signed order flow and u_t is an i.i.d. random variable. Panel C presents the estimates of the PIN model proposed in Easley et al. (1996, 1997a, 1997b). These estimates are established by maximizing the likelihood

$$L(\theta | \{B_t, S_t\}_{t=1}^T) = \prod_{t=1}^T \left((1 - \alpha) \kappa^{-\frac{B_t}{S_t}} \frac{B_t}{B_t!} e^{-\frac{B_t}{S_t}} + \alpha \kappa^{-\frac{S_t}{B_t}} \frac{S_t}{S_t!} e^{-\frac{S_t}{B_t}} \right) \frac{(\mu + \kappa)^{B_t}}{B_t!} e^{-\frac{(\mu + \kappa) B_t}{S_t}} + \alpha (1 - \beta) \kappa^{-(\mu + \kappa) \frac{B_t}{S_t}} \frac{(\mu + \kappa)^{B_t}}{B_t!} e^{-\frac{(\mu + \kappa) B_t}{S_t}} \frac{(\mu + \kappa)^{S_t}}{S_t!} e^{-\frac{(\mu + \kappa) S_t}{B_t}} \right),$$

where B_t and S_t indicate the total number of buys and sells on day t , respectively, and $\theta = (\alpha, \beta, \mu, \kappa, \tau_B, \tau_S)$ is the vector of parameters.

Panel A: PI (Price Impact) Estimates					Inf. Asym. Measure ^c	
Parameter Estimates						
	γ Price Impact Measure ^a	ϕ Gross Profit Component ^b	γ PI ^a			
A-share	Mean	9.66	85.5	9.66		
	St. Dev.	8.31	39.4	8.31		
	N sig @1%	76	76			
	N sig @1%	76	76			
B-share	Mean	2.61	63.5	2.61		
	St. Dev.	1.62	18.4	1.62		
	N sig @1%	73	76			
	N sig @1%	73	76			

(continued)

(continued)

Table II, part 1: PI model and Panel A

Panel B: AS (Adverse Selection) Estimates									
Parameter Estimates								Inf. Asym. Measure ^c	
	γ_0 Fixed AS Component ^a	γ_1 Variable AS Component ^a	γ_2 Fixed Gross- Profit Component ^b	γ_3 Variable Gross- Profit Component ^a	γ Discreteness	σ_a St. Dev. Error Term	$\gamma_0 + \gamma_1 V^*$ AS Component for Median-Size Trade ^b		
A-share	Mean	42.2	9.2	66.9	-1.2	0.18	0.020	56.4	
	St. Dev.	3.0	0.5	2.5	0.4	0.01	0.011	30.0	
	N sig @1%	76	76	76	55	76			
	N sig @1%	76	76	76	55	76			
B-share	Mean	33.5	2.1	42.5	-0.2	0.23	0.014	46.8	
	St. Dev.	8.9	0.7	8.0	0.5	0.04	0.005	21.2	
	N sig @1%	69	61	67	4	72			
	N sig @1%	69	61	67	4	72			

Panel C: PIN Model Estimates									
Parameter Estimates								Inf. Asym. Measure ^c	
	β Arrival Rate of Informed Traders	τ_B Arrival Rate of Uninformed Buys	τ_S Arrival Rate of Uninformed Sells	α Probability of an Information Event	β Probability That New Information Is Bad News			Inf. Asym. Measure ^c	
								$\frac{\alpha(\beta\mu + \tau_B + \tau_S)^{-1}}{PIN}$	
A-share	Mean	0.38	0.44	0.51	0.36	0.30	0.13		
	St. Dev.	0.14	0.16	0.17	0.07	0.22	0.04		
	N sig @1%	76	76	76	76	49			
	N sig @1%	76	76	76	76	49			
B-share	Mean	0.11	0.06	0.07	0.31	0.28	0.20		
	St. Dev.	0.04	0.03	0.03	0.08	0.23	0.04		
	N sig @1%	73	73	73	71	42			
	N sig @1%	73	73	73	71	42			

^aIn 1/10,000 yuan per 1,000 shares.

^bIn 1/10,000 yuan.

^c V^* equals the median trade size.

Table II, part 2: AS and PIN estimates

5.4 Figures 4-5 and Table III: relation among measures and discount

Read:

- Figure 4 shows a strong relationship between PI and AS, especially within the A-share and B-share markets separately.
- The AS-PIN relationship is weaker because PIN is sensitive to the level of uninformed trading.
- Figure 5 shows that B-share discounts rise with PI and AS information-asymmetry differences, but the PIN relation is weaker.
- Table III shows that the PI and AS differences are highly correlated, supporting the interpretation that both capture a similar information-asymmetry force.

Economic message: The strongest empirical measures are PI and AS. PIN is informative but less clean in this setting because it mixes informed-trading intensity with the scarcity of uninformed trades.

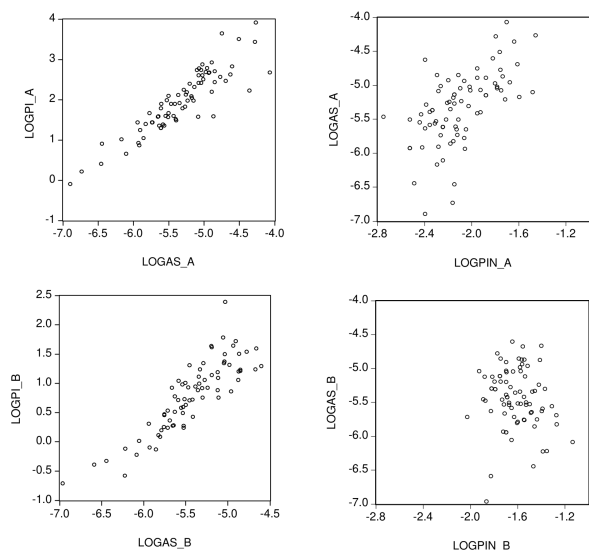


Figure 4: microstructure measures compared with each other

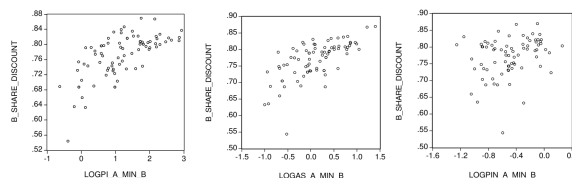


Figure 5: Scatterplots of foreign share discount versus information asymmetry mea-

Figure 5: B-share discount versus information-asymmetry measures

Table III
Correlation Explanatory Variables

is table presents the correlations among all candidate explanatory variables for the cross-sectional variation in foreign share discounts. These planatory variables include the three measures of information asymmetry differences across markets (PI, AS component of the spread, and PIN), rnover, number of trades, market capitalization in both markets, and momentum. All variables are calculated based on the period from January 00 through December 2000.

	Diff. AS Component	Diff. PIN ^a	Turnover A-share	Turnover B-share	Number of Trades A-share	Number of Trades B-share	Market Cap. A-share	Market Cap. B-share	Momentum ^b
ff PI	0.84**	0.16	0.32**	0.60**	-0.79**	0.26**	-0.69**	-0.03	0.36**
ff AS component		0.18	0.29**	0.51**	-0.67**	0.35**	-0.45**	0.07	0.20*
ff PIN			0.08	-0.05	-0.24**	-0.11	-0.20*	-0.18	0.13
rnover A-share				0.35**	-0.25**	-0.32**	-0.70**	-0.52**	0.34**
rnover B-share					-0.52**	0.26**	-0.44**	-0.27**	0.24**
umber of trades A-share						0.05	0.70**	0.11	-0.15
umber of trades B-share							0.26**	0.71**	0.07
arket cap. A-share								0.50**	-0.18
arket cap. B-share									-0.17

ocal) A-share minus (foreign) B-share.

-share return.

*Significant at the 90%/ 95% level.

Table III: correlations among information asymmetry and control variables

5.5 Table IV: main discount regressions

Read:

- In the univariate regressions, the PI difference explains about 44% of the discount variation and the AS difference explains about 46%.
- The PIN difference explains only about 8% in the univariate regression.
- With controls included, the PI and AS differences remain strongly positive and statistically significant.
- Turnover, number of trades, market capitalization, and momentum explain parts of the discount, but they do not dominate the information measures.

Economic message: The main pricing result is that direct measures of information asymmetry are more powerful than liquidity or speculative-trading controls in explaining B-share discounts.

Table IV
Foreign Share Discount: Information Asymmetry and Competing Explanations

This table presents regressions that relate the foreign share discount (FSD) to various explanatory variables. First, based on our information asymmetry hypothesis, we relate the FSD to our measures of information asymmetry, namely, the difference in PI (price impact), the AS component of the spread, and PIN. Second, we study the explanatory power of share turnover that, for the A- (local) share market, proxies for the amount of "speculative trading." According to Mei et al. (2003), this potentially leads to an A-share premium. For the B- (foreign) share market, it proxies for inactive trading which might explain B-share discounts. Third, we replace share turnover by the number of trades. Fourth, we study market capitalization as an explanatory variable. Mei et al. (2003) use it as an alternative proxy for speculative trading. Alternatively, market capitalization in the A-share market could be interpreted as a proxy for information asymmetry. Fifth, we relate the FSD to momentum, a variable that Karolyi and Li (2003) find to be important in explaining the FSD. Finally, we relate the FSD to our measures of information asymmetry, controlling for all competing explanations by adding the explanatory variables proposed by others. Turnover is defined as $\log(1 + (\text{Average Volume}/\text{Outstanding Shares}))$; market capitalization is the number of shares outstanding times the average daily closing price; momentum is the A-share return. The variables are based on January through December 2000, a period during which both markets were fully segmented. *t* values are in parentheses.

Model	Diff. PI ^a	Diff. AS Comp. ^b	Diff. PIN ^c	Turnover A-share	Turnover B-share	Number of Trades A-share	Number of Trades B-share	Market Cap. A-share	Market Cap. B-share	Momentum	Intercept	R ²
1 (PI diff)	0.045** (7.64)										0.72** (83.47)	0.44
2 (AS diff)		0.069** (7.84)									0.76** (153.36)	0.46
3 (PIN diff)			0.049** (2.58)								0.80** (71.04)	0.08
4 (Share turnover)				0.052** (2.61)	0.020* (1.77)						1.07** (13.75)	0.17
5 (Number of trades)						-0.0735** (-3.78)	0.0078 (0.44)				1.13** (9.73)	0.16
6 (Market cap.)								-0.037** (-4.34)	-0.007 (-0.98)		1.68** (10.90)	0.32
7 (Momentum)										0.042** (3.33)	0.75** (86.16)	0.13
8 (PI diff + controls)	0.065** (7.25)			0.033 (1.82)	-0.07** (-5.81)	-0.033 (-1.08)	0.198** (4.53)	0.030** (2.34)	-0.089** (-6.60)	0.010 (1.19)	1.258** (6.47)	0.77
9 (AS diff + controls)		0.079** (6.25)		-0.029 (-1.17)	-0.043** (-2.83)	0.029 (0.74)	0.079** (2.24)	-0.027* (-1.85)	-0.059** (-3.81)	0.003 (0.28)	1.69** (9.98)	0.74
10 (PIN diff + controls)			-0.006 (-0.39)	0.0375 (1.33)	-0.070** (-3.85)	-0.13** (-3.13)	0.20** (5.10)	0.006 (0.34)	-0.098** (-4.79)	0.01 (1.01)	2.28** (11.1)	0.59
11 (AS diff + PIN diff + controls)		0.080** (6.25)	0.009 (0.67)	-0.034 (-1.31)	-0.037** (-2.97)	0.043 (0.97)	0.068* (1.70)	-0.030* (-1.96)	-0.053** (-2.96)	0.003 (0.33)	1.64** (8.60)	0.75

^aA-share PI minus B-share PI.
^bA-share AS component minus B-share AS component.
^cA-share PIN minus B-share PIN.
^{*}Significant at the 90%/95% level.

Table IV: foreign share discount, information asymmetry, and competing explanations

5.6 Table V: before and after domestic investors enter the B-share market

Read:

- The March 2001 opening allows domestic investors to trade B-shares, reducing the segmentation between investor groups.
- The average B-share discount falls from about 72% before the opening to about 43% afterward.
- A-share and B-share microstructure measures become more similar after the opening.
- The AS difference between A- and B-share markets narrows substantially.

Economic message: This is the paper's institutional validation exercise. When the domestic-information advantage can enter the B-share market, the discount falls and the information gap shrinks.

Table V
PI, AS Component, and PIN: before and after Domestic Investors Are Allowed to Trade Foreign Shares
This table presents cross-sectional statistics on the parameter estimates of the econometric models we use to measure the level of information asymmetry. We present these statistics for 76 stocks for a period when the A- (local) and B- (foreign) share markets were fully segmented, as in Table II, and for a period after the B-share market was opened to domestic investors, that is, April through November 2001. Panel A presents the estimates of the price impact measure (PI), which is based on Kyle (1985) and Glosten and Harris (1988)

$$P_t - P_{t-1} = \gamma Q_t V_t + \phi(Q_t - Q_{t-1}) + e_t,$$

where P_t is the price of transaction t , Q_t is an indicator that classifies the transaction as buyer-initiated (+1) or seller initiated (-1), V_t is the transaction size, e_t is the unobservable error term, ϕ is the gross profit component associated with an order, and γ is the (inverse) market depth parameter, which is directly related to the price impact parameters introduced in the theoretical model. Panel B presents the estimates of a spread decomposition model based on Glosten and Harris (1988), defined as

$$P_t - P_{t-1} = c_0(Q_t - Q_{t-1}) + c_1(Q_t V_t - Q_{t-1} V_{t-1}) + z_0 Q_t + z_1 \delta_t + u_t \text{ with } u_t = \eta_t - \gamma \eta_{t-1},$$

where η_t is the innovation in signed order flow and u_t is an i.i.d. random variable. Panel C presents the estimates of the PIN model proposed in Easley et al. (1996, 1997a, 1997b). These estimates are established by maximizing the likelihood

$$L(\theta | \{R_t, S_t\}_{t=1}^T) = \prod_{t=1}^T \left((1 - \alpha) e^{-\gamma \frac{R_t}{B_t}} e^{-\frac{\delta}{S_t}} + \alpha e^{-\gamma \frac{R_t}{B_t}} e^{-\delta \frac{R_t}{S_t}} \right)^{\frac{R_t}{B_t}} e^{-\delta \frac{R_t}{S_t}} + \alpha (1 - \delta) e^{-(\alpha + \gamma) \frac{R_t}{B_t}} + \alpha (1 - \delta) e^{-(\alpha + \gamma) \frac{R_t}{B_t}} e^{-\delta \frac{R_t}{S_t}}.$$

where B_t and S_t indicate the total number of buys and sells on day t , respectively, and $\theta = (\alpha, \delta, \mu, \tau_B, \tau_S)$ is the vector of parameters.

Panel A: PI (Price Impact) Estimates				
Parameter Estimates				Inf. Asym. Measure
		γ	ϕ	
		Price Impact Measure ^a	Gross Profit Component ^b	PI ^a
A-share	Preevent	9.7	85.8	9.7
	Postevent	11.7	87.1	11.7
B-share	Preevent	2.6	63.5	2.6
	Postevent	4.7	57.0	4.7

(continued)

Table V—Continued

Panel B: AS (Adverse Selection) Estimates								
Parameter Estimates							Inf. Asym. Measure ^d	
		z_0	z_1	c_0	c_1	γ	σ_u	$z_0 + z_1 V^*$
		Fixed AS	Variable AS	Fixed-Gross-	Variable-Gross-	Discreteness	St. Dev.	AS Component
		Component ^b	Component ^b	Profit	Profit		Error	for Median-Size
				Component ^b	Component ^a		Term	Trade ^b
A-share	Preevent	42.2	9.2	66.9	-1.2	0.18	0.020	56.4
	Postevent	52.7	11.1	59.6	-1.7	0.15	0.020	67.4
B-share	Preevent	35.5	2.1	42.5	-0.2	0.23	0.014	46.8
	Postevent	50.9	4.2	36.3	-0.8	0.11	0.017	67.4

Panel C: PIN Model Estimates								
Parameter Estimates							Inf. Asym. Measure ^d	
		μ	τ_B	τ_S	α	δ		
		Arrival Rate	Arrival Rate	Arrival Rate	Probability of	Probability That		
		of Informed	of Uninformed	of Uninformed	an Information	New Information		
		Traders	Buyers	Sells	Event	Is Bad News		
								$\alpha/(\alpha\mu + \tau_B + \tau_S)^{-1}$
								PIN
A-share	Preevent	0.38	0.44	0.51	0.36	0.30		0.13
	Postevent	0.27	0.26	0.35	0.38	0.23		0.15
B-share	Preevent	0.11	0.06	0.07	0.31	0.28		0.20
	Postevent	0.29	0.21	0.31	0.42	0.33		0.19

^aIn 1/10,000 yuan per 1,000 shares.

^bIn 1/10,000 yuan.

^c V^* equals the median trade size.

Table V, part 1: pre- and post-opening PI estimates

Table V, part 2: pre- and post-opening AS and PIN estimates

5.7 Table VI: postevent regressions

Read:

- The postevent regressions repeat the Table IV logic after the B-share market opens to domestic investors.
- PI and AS differences remain positive and significant.
- The PI specification explains about 61% of postevent discount variation, while the AS specification explains about 71%.
- The postevent evidence supports the same mechanism even after the market structure changes.

Economic message: The postopening period does not eliminate the information-asymmetry pricing relation. Instead, it reinforces that the size of the information gap helps explain the remaining discount.

Table VI
Foreign Share Discount: Information Asymmetry and Competing Explanations, Postevent

This table relates the foreign share discount to our measures of information asymmetry and various control variables in the same spirit as Table IV. The main difference is that estimations are based on a period after the foreign share market was opened to domestic investors, April 2001 through October 2001.

	Diff. PI ^a	Diff. AS Comp. ^b	Diff. PIN ^c	Turnover A-share	Turnover B-share	Number of Trades A-share	Number of Trades B-share	Market Cap. A-share	Market Cap. B-share	Momentum	Intercept	R ²
PI diff)	0.101** (9.51)										0.31** (23.88)	0.61
AS diff)		0.213** (13.51)									0.43** (59.95)	0.71
PIN diff)			0.082* (1.94)								0.43** (25.32)	0.05
Share turnover)				0.068** (3.25)	-0.050 (-1.44)						0.51** (3.39)	0.13
Number of trades)						-0.208** (-4.78)	0.099* (1.82)				0.98** (3.49)	0.24
Market cap.)								-0.105** (-6.77)	0.089** (4.67)		0.74* (1.90)	0.41
Momentum)										0.25** (2.64)	0.46** (20.47)	0.09
PI diff + controls)	0.140** (9.17)			0.184** (7.58)	-0.214** (-7.91)	-0.253** (-4.97)	0.258** (4.17)	0.156** (6.16)	-0.163** (-6.29)	0.128** (2.23)	0.41 (1.63)	0.82
AS diff + controls)		0.266** (10.87)		-0.029 (-1.10)	-0.148** (-6.13)	0.047 (0.80)	0.181** (3.15)	-0.03 (-1.49)	-0.09** (-4.40)	0.133** (2.58)	1.11** (5.01)	0.85
(PIN diff + controls)			0.018 (0.54)	0.141** (3.78)	-0.162** (-3.87)	-0.359** (-4.37)	0.359** (3.67)	0.032 (0.94)	-0.052 (-1.46)	0.133 (1.54)	0.89** (2.45)	0.59
(AS diff + PIN diff + controls)	0.269** (11.09)	0.034 (1.65)	-0.041 (-1.49)	-0.135** (-5.42)	0.088 (1.39)	0.152** (2.55)	-0.04* (-1.96)	-0.09** (-3.95)	0.141** (2.74)	1.12** (5.15)		0.86

share PI minus B-share PI.
share AS component minus B-share AS component.
share PIN minus B-share PIN.
Significant at the 90%/95% level.

Table VI: postevent foreign share discount regressions

6 Short summary

- The paper studies whether information asymmetry affects asset prices in China's segmented A- and B-share markets.

- A- and B-shares have identical cash-flow rights but different investor clienteles.
- B-shares trade at large discounts relative to A-shares, creating a clean pricing puzzle.
- The authors build a simple model in which a fraction of domestic investors is informed while foreign investors are uninformed.
- The model predicts that A-minus-B price impact differences can be positively related to B-share discounts.
- The data are transaction and quote data for 76 dual-listed firms from January 2000 to November 2001.
- The paper estimates PI, AS, and PIN separately for A- and B-share markets.
- A-shares have higher price impact and larger adverse selection components than B-shares.
- Differences in PI and AS explain a large portion of the cross-sectional variation in B-share discounts.
- The results survive controls for trading activity, speculation, firm size, and momentum.
- After domestic investors are allowed to trade B-shares in 2001, the discount falls and the information-asymmetry gap narrows.
- The evidence supports the view that information asymmetry is priced in international equity markets.

7 Conclusion

7.1 Core conclusion

Chan, Menkveld, and Yang show that information asymmetry between domestic and foreign investors helps explain the China foreign-share discount. Microstructure measures of informed trading - especially price impact and the adverse selection component of the spread - are strongly related to the cross-section of B-share discounts.

7.2 Economic intuition

Domestic investors have better access to local information, rumors, and market conditions. When this domestic informational advantage is large, A-share prices are higher relative to B-share prices. Market microstructure measures reveal where informed trading is most severe, and those same stocks exhibit larger foreign-share discounts.

7.3 Takeaway

The paper connects market microstructure to asset pricing. The same forces that make trades more informationally costly in one market also help determine the relative prices of otherwise identical claims across segmented investor groups.